

Army pool sensitivity — run matrix (for dummies)

Goal: See how pool grain, TB-zero exclusion, and straight LOO vs minus-mean change H_sort (panel 5) and HERO porch (panel 9). Always work from AWS 520 root (`Network_1P_shell/` — folder with `big_dfs/` , `520_pipeline_cox_working.ipynb` , `talent/re_entry/`). Vocabulary (one time):

Plain English	Config / column
Pool grain = LEGACY	<code>POOL_GROUPING_MODE = 'legacy'</code> — peers = same snapshot x same SNR rater
Pool grain = THRU	<code>POOL_GROUPING_MODE = 'active_at_eval_thru'</code> — peers active at OER eval thru date
Keep TB zeros in pool	<code>POOL_EXCLUDE_PEER_TB_ZERO = False</code>
Drop TB-zero peers from pool mean	<code>POOL_EXCLUDE_PEER_TB_ZERO = True</code>
Straight LOO (T_j) — basketball-style pond level	Column <code>pool_tb_ratio_mean_snr_fwd</code>
Minus-mean — player minus team mean	Column <code>pool_minus_mean_snr_fwd</code> / HERO <code>z_pool_minus_mean_snr_fwd</code>

Five runs (3 notebook + 2 plot-only):

Run	Grain	TB zeros in pool?	Peer X in plots	Notebook re-run?
1 · LEGACY	legacy	keep	straight LOO	Yes
2 · THRU	active_at_eval_thru	keep	straight LOO	Yes
3 · THRU_NOZERO	active_at_eval_thru	exclude	straight LOO	Yes
4 · LEGACY_MINUS (optional)	legacy	keep	minus-mean	No — reuse Run 1 feather
5 · THRU_MINUS (optional)	thru	keep	minus-mean	No — reuse Run 2 feather

What to compare for Alex (PD41): Panel 5 title line (`H_sort=...`) and panel 9 porch shape. Write numbers in the table at the bottom of this file.

RUN 1 · LEGACY · straight LOO · keep TB zeros

What this run means: Old pool definition + peer pond level (not minus-mean). Baseline reference.

1a · RUN these commands in the terminal (AWS)

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big_dfs/df_pipeline_11_cox_analysis.feather
```

(The last line is optional — just confirms whether an old feather exists before you rebuild it in step 1d.)

1b · EDIT `pipeline_config.py` at 520 root, then SAVE

Change these lines to these exact values:

```
POOL_GROUPING_MODE = 'legacy'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

1c · EDIT two plot scripts on AWS, then SAVE (do not run anything yet)

Purpose: Tell the plot code to use straight LOO pond level, not minus-mean and not the missing z-column. EDIT file 1: open `talent/re_entry/army_basic_plots.py` in JupyterLab. Go to line 42. Change the line so it reads exactly:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

Save the file.

EDIT file 2: open `talent/re_entry/army_hero_slide_plot.py`.

Go to **line 44**.

Change the line so it reads exactly:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

Save the file.

SKIP file 3: do not edit `army_act2_probes.py` for Run 1.

Do not set `plot_var` to `z_pool_tb_ratio_mean_snr_fwd` — that column is not in the feather unless you later run Path B (Cell 10.5 with a config change).

Nothing to check in this step — only edit and save the two files above.

1d • RUN notebook cells in this order (JupyterLab)

Open `520_pipeline_cox_working.ipynb`.

RUN **Cell 0** (reloads config from step 1b).

RUN **Cell 5** (pool means — in the log, confirm you see `POOL_GROUPING_MODE='legacy'`).

RUN **Cell 6** (pass-through is OK) **OR** if Cell 6 errors, RUN this in the terminal instead:

```
cp big_dfs/df_pipeline_05_pool_means.feather big_dfs/df_pipeline_06_pool_ranks.feather
```

RUN **Cell 7**.

RUN **Cell 8**.

RUN **Cell 9**.

RUN **Cell 10**.

RUN **Cell 10.5**.

RUN **Cell 11** (writes `big_dfs/df_pipeline_11_cox_analysis.feather`).

1e • RUN this check in the terminal (after Cell 11)

```
python3 - <<'PY'
import pandas as pd
df = pd.read_feather("big_dfs/df_pipeline_11_cox_analysis.feather")
for c in ["pool_tb_ratio_mean_snr_fwd", "pool_minus_mean_snr_fwd", "z_pool_minus_mean_snr_fwd"]:
    print(c, "YES" if c in df.columns else "MISSING")
PY
```

Expect `pool_tb_ratio_mean_snr_fwd: YES`.

1f • RUN these plot commands in the terminal, then EDIT the manifest, then RUN mosaic

RUN:

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
```

EDIT `manifests/army_run1_3x3_manifest.json`: set panel 9 `"path"` to exactly:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

RUN:

```
python talent/re_entry/build_army_data_story.py
```

1g • RUN these checks — did Run 1 work?

```
grep POOL_GROUPING_MODE pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the **BDP log**, find the line:

```
Overlap pool grain: POOL_GROUPING_MODE='legacy'
```

Open panel 5 PNG — note **H_sort** in the title.

Open panel 9 — note porch shape.

1h • RUN this backup command before you start Run 2

Also RUN (saves feather for optional Run 4 later):

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run1_legacy.feather
```

```
bash scripts/backup_rename_suffix.sh _run1_legacy --all-output
```

RUN 2 • THRU • straight LOO • keep TB zeros

What this run means: Same peer metric as Run 1, but **eval-thru pool grain** — isolates **grain / assortativity** effect.

2a • Before you start

```
cd /path/to/Network_1P_shell
```

2b • `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_eval_thru'  
POOL_EXCLUDE_PEER_TB_ZERO = False  
CELL5_POOL_MEANS = True  
CELL6_POOL_RANKS = True
```

Save.

2c • Plot scripts

Same as Run 1 — straight LOO (`pool_tb_ratio_mean_snr_fwd`).

No change if you already set Run 1 values.

2d • Notebook cells

Same order as Run 1: **0 → 5 → 6 (or cp) → 7 → 8 → 9 → 10 → 10.5 → 11**.

Watch Cell 5 log for `POOL_GROUPING_MODE='active_at_eval_thru'`.

2e • Feather sanity check

Same probe as Run 1e.

2f • CLI

Same as Run 1f.

Manifest panel 9 path unchanged (same HERO filename).

2g • Checks

```
grep POOL_GROUPING_MODE pipeline_config.py
```

BDP log must show:

```
Overlap pool grain: POOL_GROUPING_MODE='active_at_eval_thru'
```

Compare **H_sort** and **HERO porch** to Run 1 backup in `output/*/run1_legacy/`.

2h • Save before Run 3

```
bash scripts/backup_rename_suffix.sh _run2_thru --all-output
```

RUN 3 • THRU • straight LOO • exclude TB-zero peers

What this run means: Same grain as Run 2, but peers with **tb_ratio == 0** dropped from pool mean/size.

3a • Before you start

```
cd /path/to/Network_1P_shell
```

3b • `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_eval_thru'  
POOL_EXCLUDE_PEER_TB_ZERO = True  
CELL5_POOL_MEANS = True  
CELL6_POOL_RANKS = True
```

Save.

3c • Plot scripts

Same as Run 1 — straight LOO. No change.

3d • Notebook cells

Same order: 0 → 5 → 6 (or cp) → 7 → 8 → 9 → 10 → 10.5 → 11.

Cell 5 log should mention `POOL_EXCLUDE_PEER_TB_ZERO=True` if enabled in code.

3e • Feather sanity check

Same as Run 1e.

Optional:

```
python -u talent/re_entry/army_pool_size_probe.py
```

3f • CLI

Same as Run 1f.

3g • Checks

Compare **H_sort** and **HERO** to Run 2 backup (`_run2_thru`).

3h • Save

```
bash scripts/backup_rename_suffix.sh _run3_thru_nozero --all-output
```

RUN 4 • LEGACY • minus-mean (optional • plot-only)

What this run means: Old relative standing peer X on the **same feather as Run 1**. No notebook.

4a • Before you start

Restore Run 1 feather **OR** never overwrite it after Run 1 — you need legacy-grain feather from Run 1.

If you already re-ran notebook for Run 2+, re-do Run 1 notebook first **or** keep a copy:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run1_legacy.feather
```

4b • `pipeline_config.py`

No change required for plots (feather already built).

4c • Plot scripts — minus-mean

`army_basic_plots.py` line 42:

```
COL_L00 = "pool_minus_mean_snr_fwd"
```

`army_hero_slide_plot.py` line 44:

```
plot_var = "z_pool_minus_mean_snr_fwd"
```

Save.

4d • Notebook

Skip — use Run 1 feather.

If feather was overwritten, copy back:

```
cp big_dfs/df_pipeline_11_run1_legacy.feather big_dfs/df_pipeline_11_cox_analysis.feather
```

4e • Feather check

```
python3 - <<'PY'
import pandas as pd
df = pd.read_feather("big_dfs/df_pipeline_11_cox_analysis.feather")
print("z_pool_minus_mean_snr_fwd", "YES" if "z_pool_minus_mean_snr_fwd" in df.columns else "MISSING")
PY
```

4f • CLI

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd
```

Manifest panel 9:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

```
python talent/re_entry/build_army_data_story.py
```

4g • Checks

Panel 5 **H_sort** should match Run 1 (same grain — only peer X axis changed on panels 3/7/8/9).

Panel 9 porch may look **different** from Run 1 even though grain is the same.

4h • Save

```
bash scripts/backup_rename_suffix.sh _run4_legacy_minusmean --all-output
```

RUN 5 • THRU • minus-mean (optional • plot-only)

What this run means: Minus-mean peer X on Run 2 thru feather.

5a • Restore Run 2 feather

```
cp big_dfs/df_pipeline_11_run2_thru.feather big_dfs/df_pipeline_11_cox_analysis.feather
```

(Save Run 2 feather right after Run 2 notebook if you have not already.)

5b • `pipeline_config.py`

No notebook change.

5c • Plot scripts

Same minus-mean settings as Run 4c.

5d • Notebook

Skip.

5e • Feather check

Same as Run 4e.

5f · CLI

Same as Run 4f.

5g · Checks

Panel 5 **H_sort** should match Run 2.
Compare panel 9 to Run 4 — shows minus-mean vs straight LOO at **thru** grain.

5h · Save

```
bash scripts/backup_rename_suffix.sh _run5_thru_minusmean --all-output
```

After all runs — fill this in for Alex

Run	Grain	TB zero excl?	Peer X	H_sort (panel 5)	HERO porch note
1 legacy	legacy	no	straight LOO		
2 thru	active_at_eval_thru	no	straight LOO		
3 thru nozero	active_at_eval_thru	yes	straight LOO		
4 legacy minus	legacy	no	minus-mean		
5 thru minus	active_at_eval_thru	no	minus-mean		

Tip: Save feathers when notebook finishes so plot-only runs work:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run1_legacy.feather # after Run 1
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run2_thru.feather # after Run 2
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run3_thru_nozero.feather # after Run 3
```

Common mistakes (read when tired)

- HERO fails with `Column not in feather: z_pool_tb_ratio_mean_snr_fwd` → use `pool_tb_ratio_mean_snr_fwd` for straight LOO (Runs 1–3).
- Mosaic fails `FileNotFoundError` → manifest panel 9 path must **exactly match** HERO output filename.
- Changed `pipeline_config.py` but **did not re-run Cell 5→11** → feather still old grain.
- Cell 0 loads `pipeline_config_div_name` → OK (it imports `pipeline_config`).
- Straight LOO scripts + legacy feather = **valid**; thru feather + minus-mean manifest = **broken names** only.